

YONATAN AWLACHEW FIRDE

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PROFILE SUMMARY

Final-year Computer Science student with a strong foundation in backend development, software engineering, and applied data analysis. Hands-on experience building full-stack web applications (ASP.NET Core, React, FastAPI) with REST API design, JWT authentication, and relational databases, alongside a solid background in data validation, machine learning, and research methodology. Currently completing an Erasmus+ exchange in Software Engineering and Web Programming at Warsaw University of Technology. Seeking a junior backend developer position to apply and grow these skills on production-grade systems.

EDUCATION

University of Messina, Messina, Italy

Bachelor's in Data Analysis | Sep 2023 – Dec 2026

- Current GPA: 29.74/30
- Coursework: Python Programming, Advanced Data Analysis, AI Applications, Data Visualization, Statistical Modeling, Operating Systems

Warsaw University of Technology, Warsaw, Poland

Erasmus+ Exchange Student – Computer Science / Data Analysis | Sep 2025 – Feb 2026

- Coursework: Software Engineering, Web Programming, Operating Systems, IoT Systems
- Focus: Full-stack application development, data-driven systems, UML Design

University of Malaya, Kuala Lumpur, Malaysia

Erasmus+ Exchange Student – Computer Science / Data Analysis | Mar 2026 – Aug 2026

- Coursework: Computational Thinking and Interactive Design
- Focus: Full-stack application development, data-driven systems, and anomaly detection research

University of Rijeka, Rijeka, Croatia

Erasmus+ Exchange Student – Computer Science / Data Analysis | Oct 2024 – Feb 2025

- Coursework: Databases, OOP, Machine Learning & AI, Data Mining
- Focus: CRUD, data-driven systems, and anomaly detection research

TECHNICAL SKILLS

Backend & Web Development: ASP.NET Core (C#), FastAPI (Python), REST API Design, JWT Authentication, Entity Framework Core

Frontend: React, JavaScript, HTML/CSS, Bootstrap

Databases: MySQL, SQL Server, SQL

Programming Languages: Python, Java, C#, JavaScript, SQL

AI & Data Tools: OpenAI API, Groq, Pandas, NumPy, Scikit-learn

Tools & Practices: Git/GitHub, Docker, Swagger/OpenAPI, Unit Testing (pytest)

Data Skills: Data Cleaning, Annotation, Quality Validation, Statistical Analysis, Visualization (Tableau)

Soft Skills: Analytical Thinking, Attention to Detail, Remote Collaboration

Languages: English (Professional), Italian (Elementary), Amharic (Native)

PROJECTS

PDF Q&A Assistant — Retrieval-Augmented Generation (RAG) API

- Built a RAG API that answers natural-language questions about PDF documents and returns answers with source citations

- Combined Gemini embeddings for semantic document retrieval with Groq for fast LLM inference, exposed through a FastAPI service
- Wrote a pytest test suite covering the retrieval and answer-generation pipeline

Technologies: Python, FastAPI, Groq, Google Gemini Embeddings, pytest

Cinema Ticket Booking System — Full-Stack Web Application

- Designed and built a full-stack seat-booking platform: an ASP.NET Core 9 Web API with JWT authentication and role-based access control, paired with a React 19 single-page frontend
- Modeled the relational schema in Entity Framework Core / MySQL and implemented seat-level reservations with database-level double-booking prevention via a unique constraint
- Added optimistic concurrency control (row-versioning) so simultaneous profile edits fail safely instead of silently overwriting each other

Technologies: ASP.NET Core, C#, Entity Framework Core, MySQL, React, JWT, Swagger/OpenAPI

Data Mining Project — Italian Agricultural Economy

- Developed a machine learning framework to analyze 43+ years of Italy's agricultural economic data (1980–2023) from EUROSTAT, identifying long-term sustainability trends
- Implemented and compared 4 classification algorithms (KNN, Naïve Bayes, SVM, Logistic Regression), achieving 85%+ accuracy in economic pattern prediction
- Applied unsupervised learning (K-Means, Hierarchical Clustering) to discover hidden economic structures and segment agricultural production factors

Technologies: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Statistical Analysis

Data Cleaning & Visualization — Global Superstore Dataset

- Cleaned and analyzed a multi-year (2011–2014) global retail dataset spanning 147 countries using Python for data cleaning and SQL for querying sales and profitability trends
- Built an interactive Tableau dashboard to surface key business insights for stakeholders

Technologies: Python, SQL, Tableau, Pandas

RELEVANT EXPERIENCE

AI Data Annotation Specialist (Freelance)

Remote | 2024 – 2025

- Labeled, categorized, and validated large datasets for AI training models
- Developed and refined data labeling workflows, ensuring 100% accuracy
- Collaborated remotely with cross-functional international teams
- Applied automation techniques using Python for quality verification and data consistency

CERTIFICATIONS & ACHIEVEMENTS

Hack4Tourism Messina 2023 – Winner (100+ competitors)

Python Programming Certification

Elements of AI & Machine Learning Certification

Palantir Foundry Foundations Certification

RESEARCH INTERESTS

- Anomaly Detection in IoT Systems (aligned with current WUT coursework)
- Machine Learning Applications in Data Quality and Validation
- AI-Driven Data Analysis and Pattern Recognition